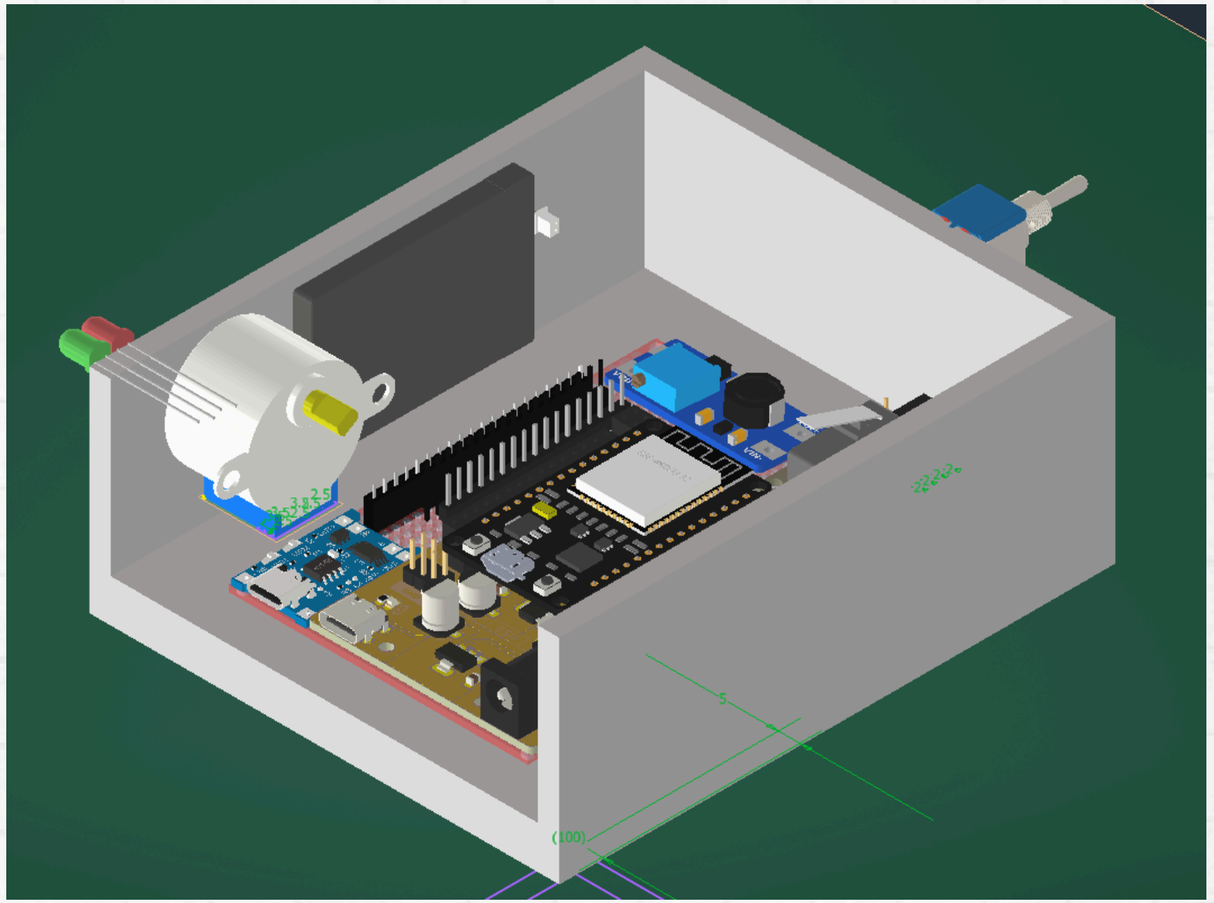
Entregable 10

Equipo 10



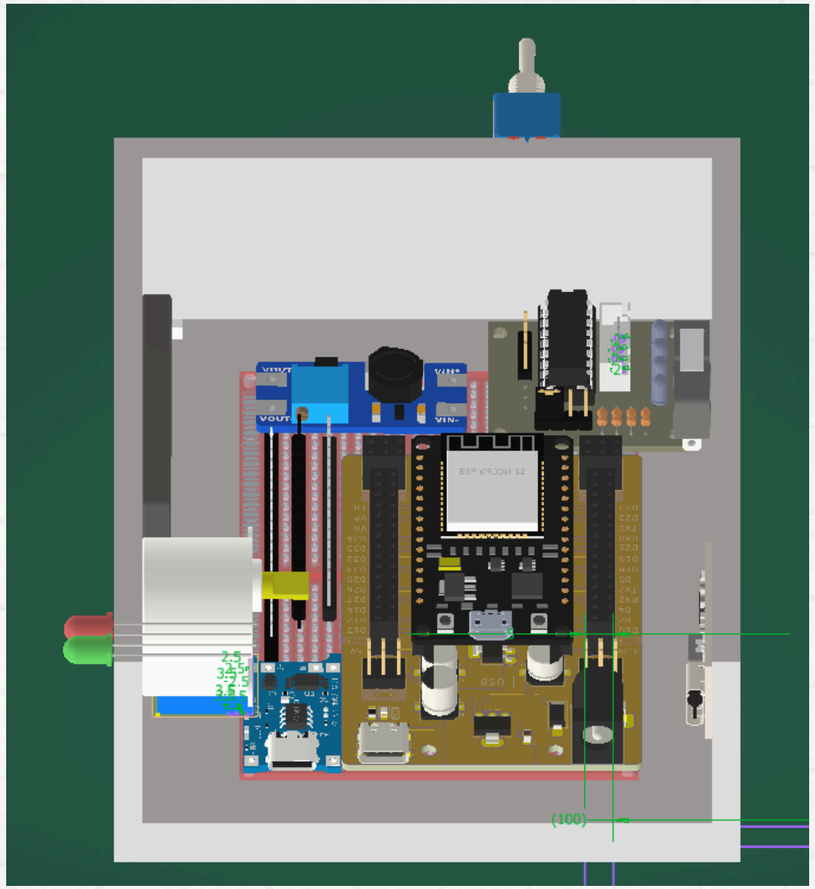
MECÁNICA Y FABRICACIÓN



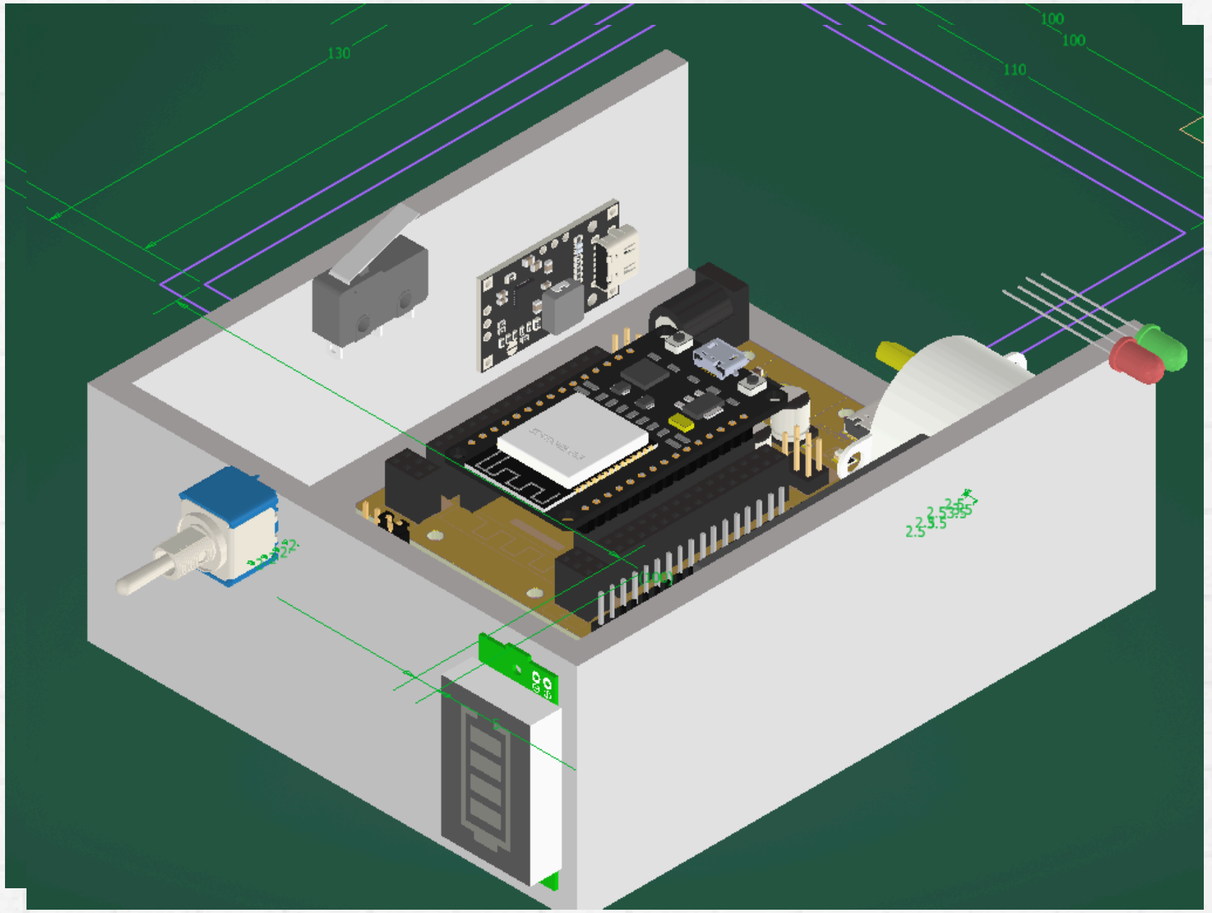
[1] "Toggle Switch 3 Positions", GrabCAD Community Library, s. f. Disponible en: https://grabcad.com/library/mts-102-mts-202-miniature-toggle-switches-on-on-1

[2] "ULN2003 Stepper Motor Driver Board", GrabCAD Community Library, s. f. Disponible en: https://grabcad.com/library/uln2003-stepper-motor-driver-board-1.

[3] "28BYJ-48 Stepper Motor", GrabCAD Community Library, s. f. Disponible en: https://grabcad.com/library/28byj-48-stepper-motor-1.

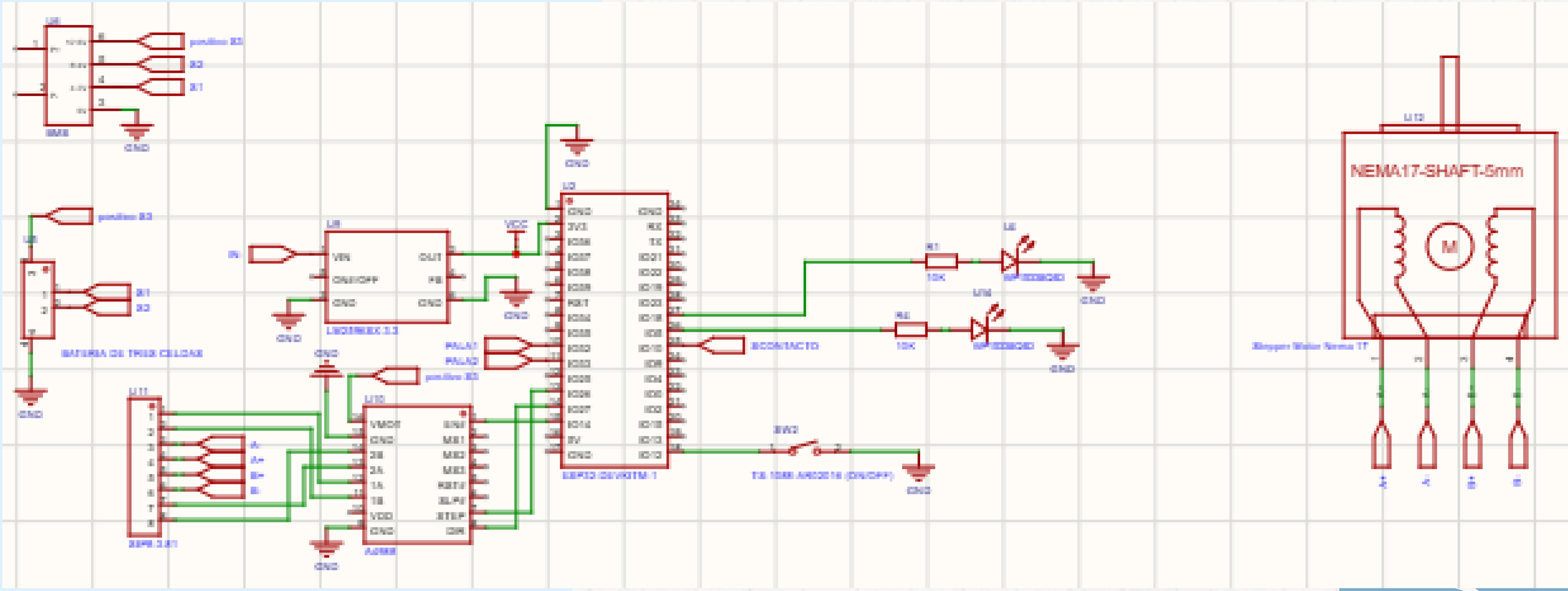


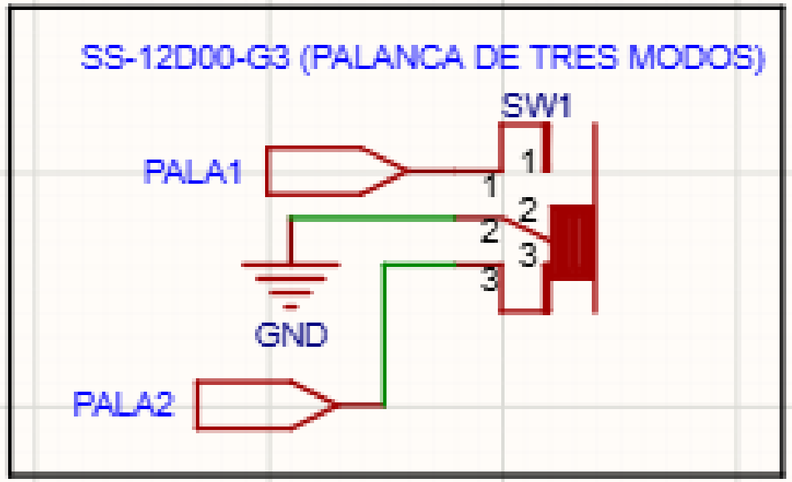
[4] "TP4056 Battery Charger Module", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/tp4056-2
[5] "Pin Header 2.54mm", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/2-54mm-pin-headers-1
[6] "Battery Capacity Indicator Display Module", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/18650-li-ion-battery-capacity-indicator-module-1
[7] "ESP32 Development Board", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/esp-32-wroom-30-pin-1
[8] "Buck Boost Converter Module", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/mt3608-holedis-7-62mm-t1-6mm-1

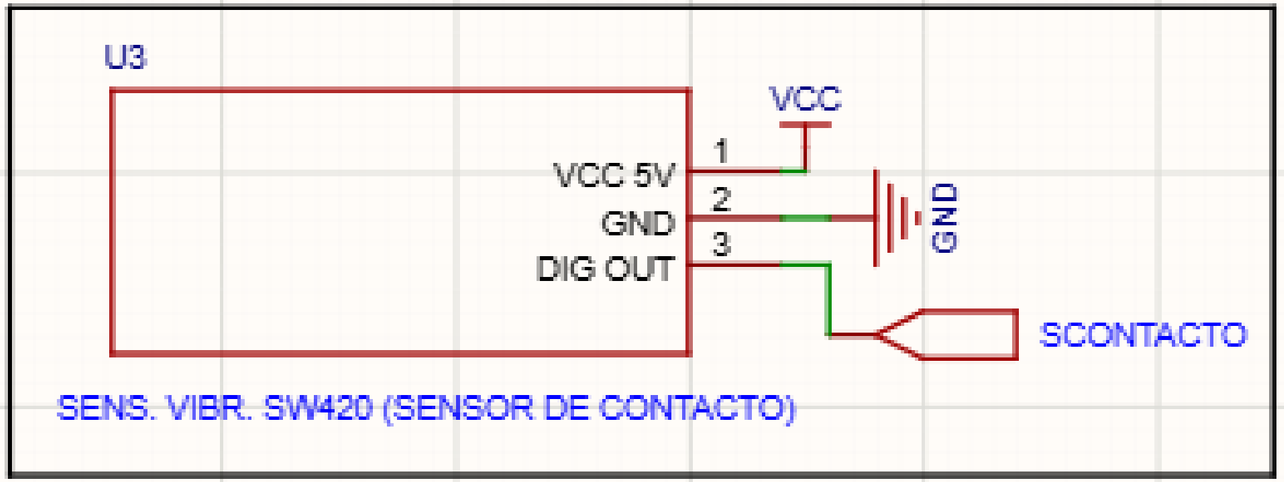


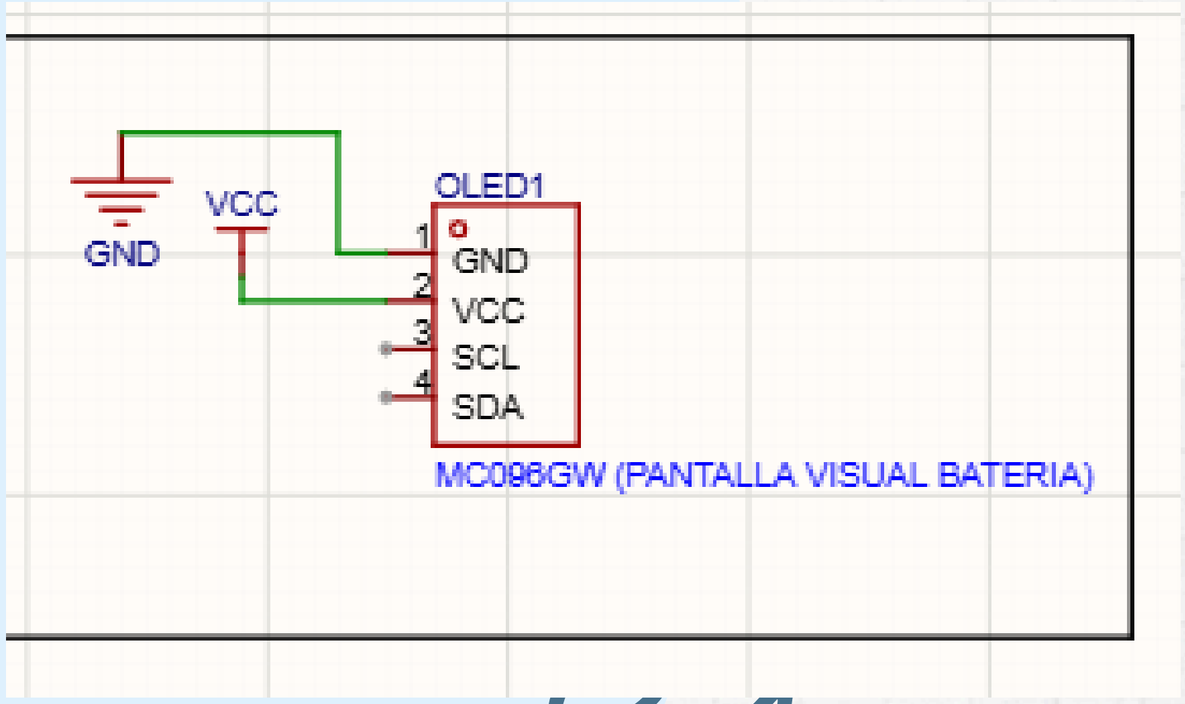
[9] "Micro Switch Limit Switch", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/kw10-z1p-micro-limit-switch-1
[10] "1S LiPo Battery 3.7V 1000mAh", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/bateria-lipo-3-7v-1000ma-503450-1/files
[11] "Oval Rocker Switch KCD1", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/kcd1-105-rocker-switch-with-light-3-pin-1
[12] "5mm LED Diodes", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/light-emitting-diode-5mm-1
[13] "Universal Prototype PCB Board 50x70mm", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/prototyping-matrix-pcb-50x70-1

ELECTRÓNICA Y SENSORES

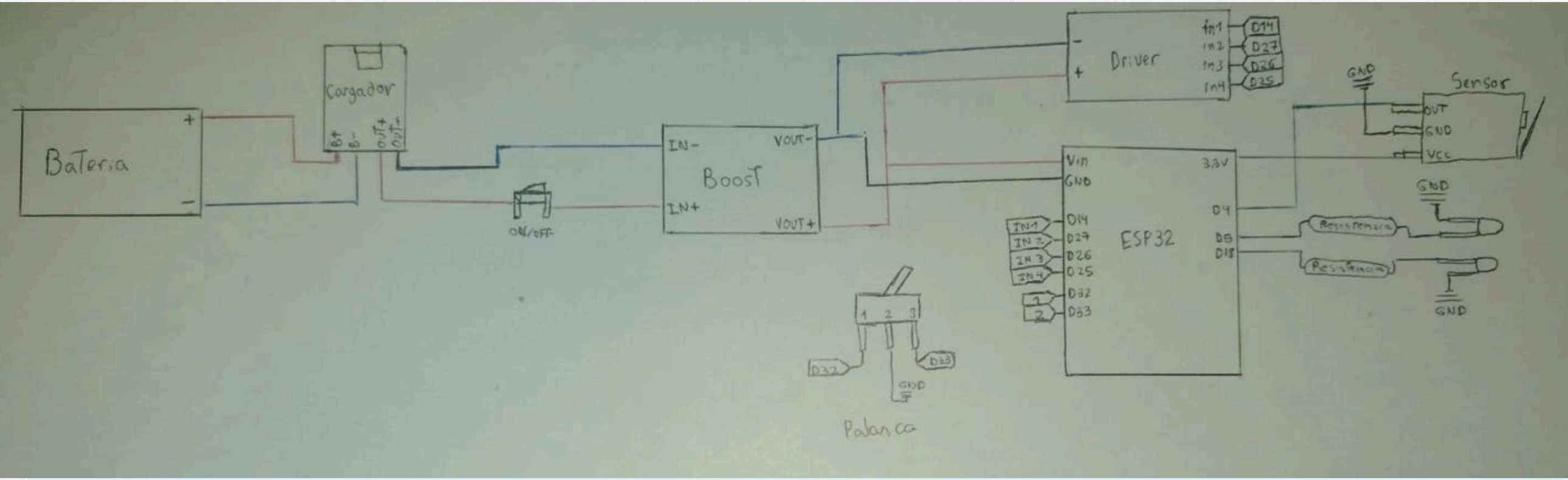



SS-12D00-G3 (PALANCA DE TRES MODOS)
SW1
PALA1
1
2
3
GND
PALA2

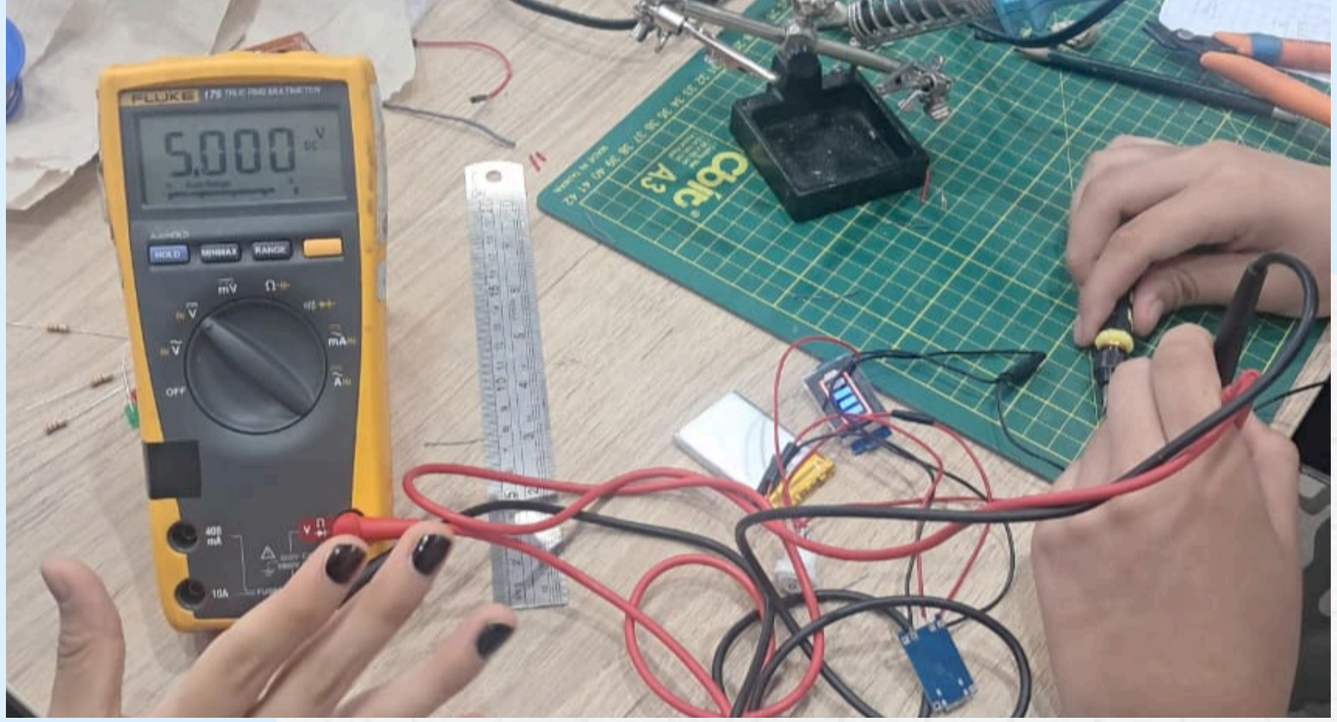

U3
VCC
VCC 5V
GND
DIG OUT
1
2
3
GND
SCONTACTO
SENS. VIBR. SW420 (SENSOR DE CONTACTO)

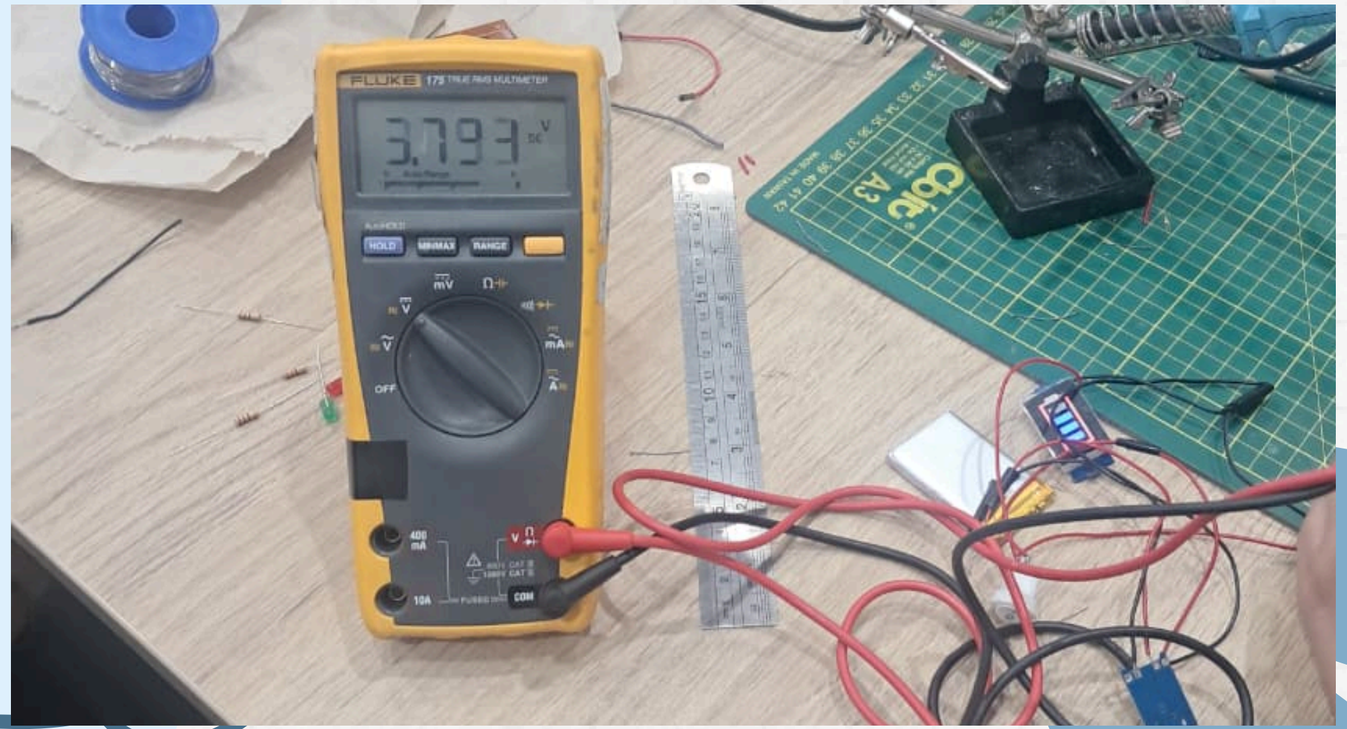

GND
VCC
OLED1
1
GND
2
VCC
3
SCL
4
SDA
MC096GW (PANTALLA VISUAL BATERIA)

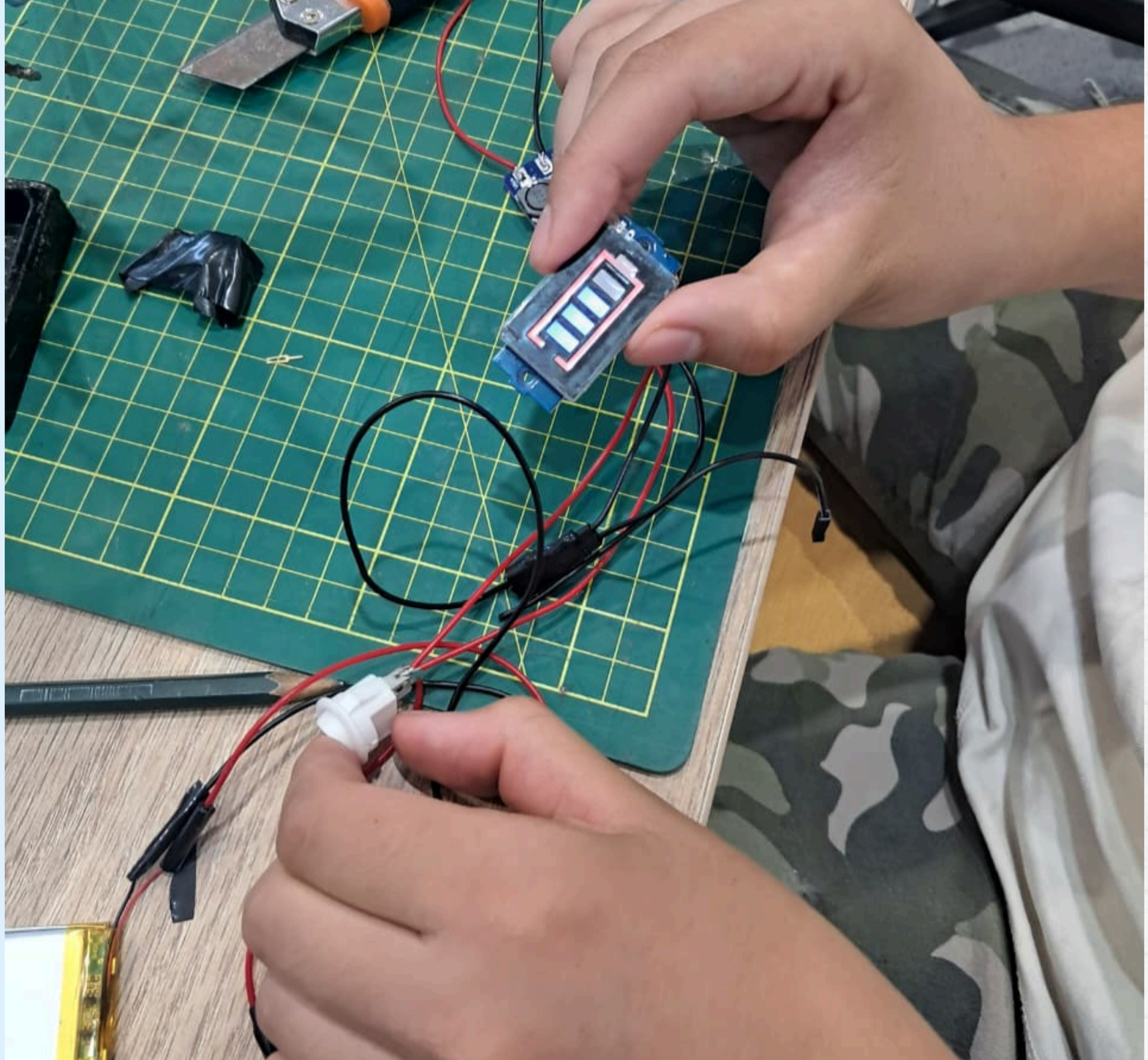
Representación de soldadura :

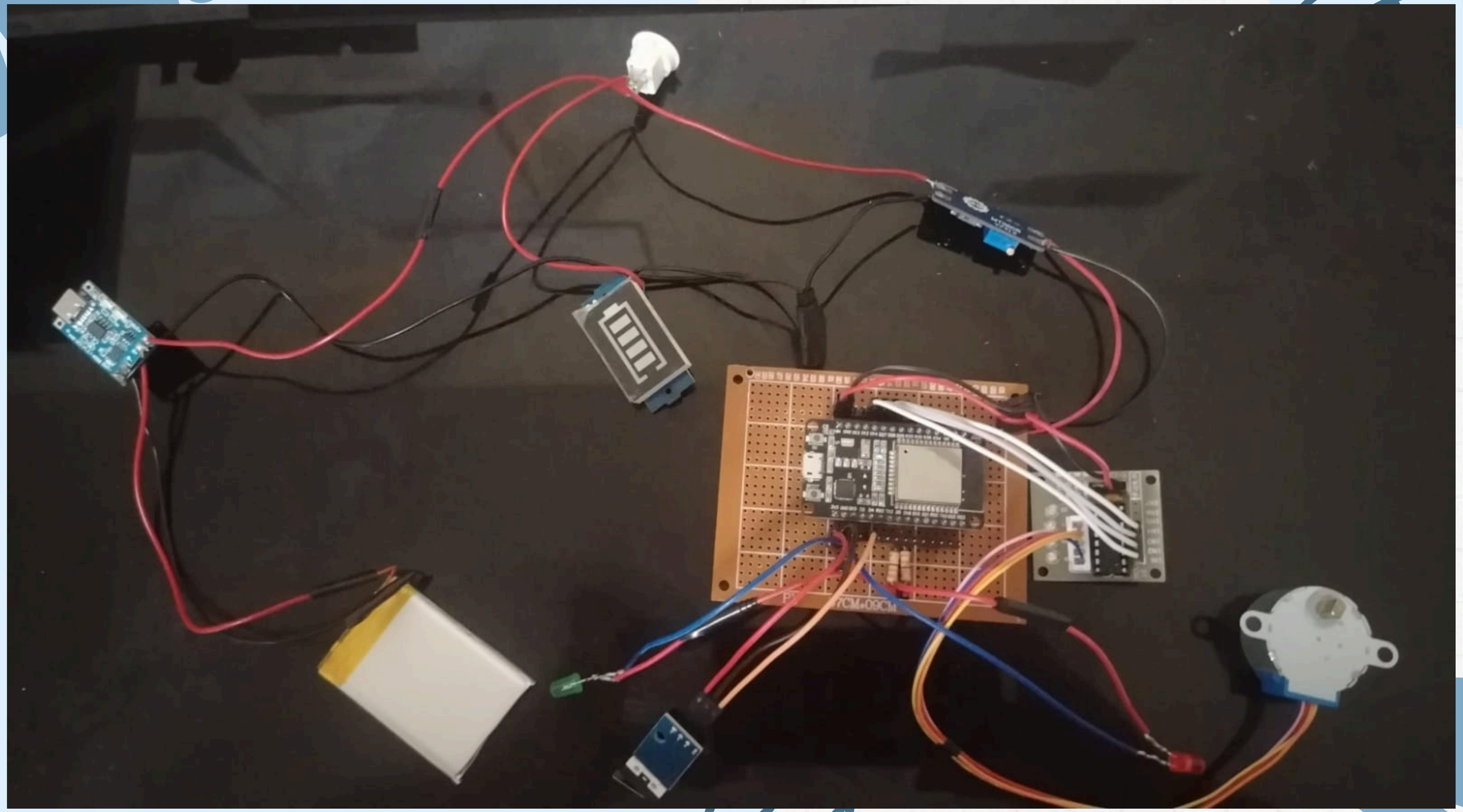


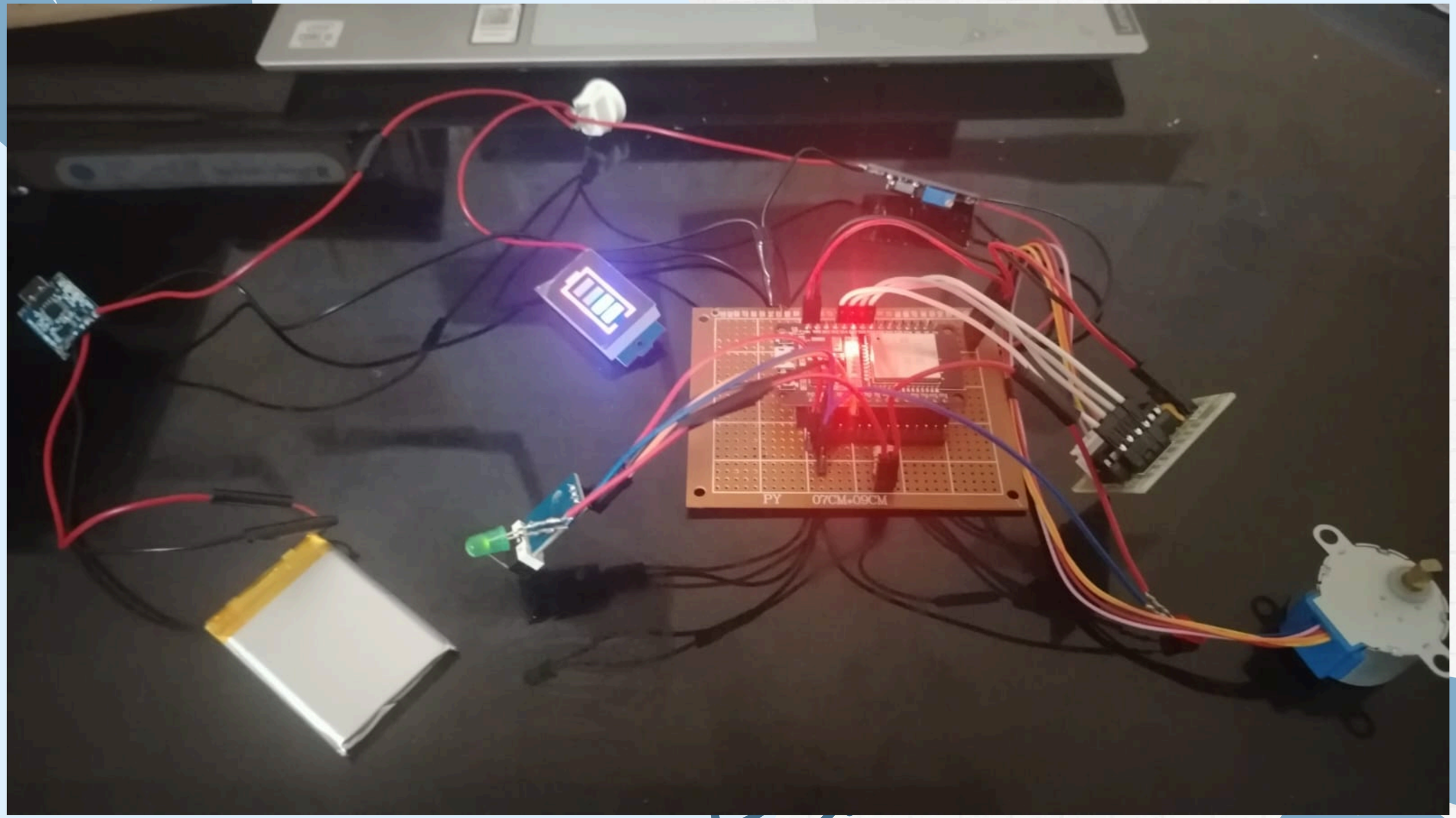
Componente	Voltaje
Batería 1S	3.7V
Convertidor MT3608.	5V
Pines de ESP32	3.3V










PY 07CM*09CM

PROGRAMACIÓN Y CONTROL

Etapa	Funcionamiento
1. Inicialización	Configuración del ESP32, pines y velocidad del motor (10 RPM).
2. Lectura de batería	Medición y visualización del voltaje mediante el GPIO34.
3. Detección de contacto	El sensor (GPIO4) habilita el funcionamiento y activa el LED verde.
4. Selección del movimiento	La palanca determina el movimiento: avance (GPIO33), retroceso (GPIO32) o detención.
5. Accionamiento	El motor 28BYJ-48 ejecuta 2048 pasos por orden (una revolución).
6. Transmisión mecánica	Los rodillos y el sistema de guiado transmiten el movimiento al catéter.

```ino
#include <Stepper.h>

const int stepsPerRevolution = 2048;
Stepper myStepper(stepsPerRevolution, 14, 27, 26, 25);

const int pinLedRojo = 18;
const int pinLedVerde = 5;
const int pinSensorContacto = 4;

const int pinModoAvance = 33;
const int pinModoRetroceso = 32;

const int pinLecturaBateria = 34;

void setup() {
  Serial.begin(115200);

  pinMode(pinLedRojo, OUTPUT);
  pinMode(pinLedVerde, OUTPUT);

  pinMode(pinSensorContacto, INPUT_PULLUP);
  pinMode(pinModoAvance, INPUT_PULLUP);
  pinMode(pinModoRetroceso, INPUT_PULLUP);
  pinMode(pinLecturaBateria, INPUT);
```

PROGRAMACIÓN Y CONTROL

```
  myStepper.setSpeed(10);

  digitalWrite(pinLedRojo, HIGH);
  digitalWrite(pinLedVerde, LOW);

  Serial.println("Sistema encendido físicamente.");
}

void loop() {
  int valorBateriaRaw = analogRead(pinLecturaBateria);
  float voltajeBateria = (valorBateriaRaw / 4095.0) * 3.3 * 2;
  Serial.print("Nivel de Bateria (ADC): ");
  Serial.print(valorBateriaRaw);
  Serial.print(" | Voltaje aprox: ");
  Serial.print(voltajeBateria);
  Serial.println(" V");

  bool pacienteDetectado = (digitalRead(pinSensorContacto) == LOW);
```

```
  if (pacienteDetectado) {
    digitalWrite(pinLedVerde, HIGH);

    bool avanza = (digitalRead(pinModoAvance) == LOW);
    bool retrocede = (digitalRead(pinModoRetroceso) == LOW);

    if (avanza && !retrocede) {
      Serial.println("Modo: Avanzando catéter (1 vuelta)...");
      myStepper.step(stepsPerRevolution);
    }
    else if (retrocede && !avanza) {
      Serial.println("Modo: Retrocediendo catéter (1 vuelta)...");
      myStepper.step(-stepsPerRevolution);
    }
    else {
      Serial.println("Modo: Detenido.");
    }

  } else {
    digitalWrite(pinLedVerde, LOW);
    Serial.println("Esperando posicionamiento en el sensor de contacto...");
    delay(500);
  }

  delay(100);
}
```

Referencias:

[1] "Toggle Switch 3 Positions", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/mts-102-mts-202-miniature-toggle-switches-on-on-1

[2] "ULN2003 Stepper Motor Driver Board", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/uln2003-stepper-motor-driver-board-1. [Accedido: 03-nov-2026].

[3] "28BYJ-48 Stepper Motor", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/28byj-48-stepper-motor-1. [Accedido: 03-nov-2026].

[4] "TP4056 Battery Charger Module", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/tp4056-2

[5] "Pin Header 2.54mm", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/2-54mm-pin-headers-1

[6] "Battery Capacity Indicator Display Module", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/18650-li-ion-battery-capacity-indicator-module-1

[7] "ESP32 Development Board", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/esp-32-wroom-30-pin-1

[8] "Buck Boost Converter Module", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/mt3608-holedis-7-62mm-t1-6mm-1

[9] "Micro Switch Limit Switch", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/kw10-z1p-micro-limit-switch-1

[10] "1S LiPo Battery 3.7V 1000mAh", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/bateria-lipo-3-7v-1000ma-503450-1/files

[11] "Oval Rocker Switch KCD1", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/kcd1-105-rocker-switch-with-light-3-pin-1

[12] "5mm LED Diodes", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/light-emitting-diode-5mm-1

[13] "Universal Prototype PCB Board 50x70mm", *GrabCAD Community Library*, s. f. Disponible en: https://grabcad.com/library/prototyping-matrix-pcb-50x70-1

Muchas Gracias

CAYETANO ES MÁS QUE CAYETANO

